Zachary H. Lemmon
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Education

University of Wisconsin – *Madison, WI* Ph.D. 2014
Laboratory of Genetics

University of Notre Dame – *Notre Dame, IN* B.S. 2008
Department of Biological Sciences

Professional Experience

Senior Computational Breeding Scientist – Computational Breeding April 2021 – January 2026
Inari Agriculture, Cambridge MA

- Developed **genomic prediction models** in R leveraging genotypic, enviromic, and phenotypic datasets to drive product advancement and accelerate genetic gain
- Built, deployed, and operated multiple cloud-based **Argo and Databricks workflows** supporting breeding programs across multiple environments and germplasm pools
- Initiated and directed development of an AWS Athena **genotype database** managing storage and usage of VCF data in genomic prediction pipelines
- Provided **line management and technical mentorship** to computational bioinformatic scientists

Bioinformatics Lead – Science Platform March 2018 – March 2021
Inari Agriculture, Cambridge MA

- **Led an agile multi-disciplinary team** of bioinformatic, molecular biology, and automation scientists to develop proprietary genome editing technology
- Implemented scalable **AWS bioinformatics workflows** for CRISPR guide RNA design and various NGS methods: plasmid, amplicon, whole genome, and bisulfite sequencing
- Established and championed software engineering best practices such as **CI/CD, version control, and testing frameworks** in the bioinformatics team
- Contributed to **IP generation** through designing novel assay methods and gene-trait associations

Postdoctoral Research Fellow July 2014 – Feb 2018
Cold Spring Harbor Laboratory, Lippman Laboratory, Cold Spring Harbor NY
Advisor: Dr. Zachary Lippman

- Developed bioinformatic analyses of **Solanaceae transcriptomes** to explore the role of stem cell maturation on phenotypic traits related to productivity
- Applied **genome editing** to exploit engineered allele combinations for improving yield

Graduate Research Associate August 2008 – August 2014
University of Wisconsin, Laboratory of Genetics, Madison WI
Dissertation: The Complex Inheritance of Maize Domestication Traits and Gene Expression
Advisor: Dr. John Doebley

- Implemented a large-scale **allele specific RNA-seq** method to explore the evolution of *cis* and *trans* gene regulation in maize
- Conducted **QTL mapping** for various domestication traits in a maize-teosinte RIL population

Undergraduate Research Assistant May 2007 – August 2008
University of Notre Dame, Department of Biology, Notre Dame IN
Advisor: Dr. Joseph O'Tousa
- Characterization of an apoptosis inducing factor in *Drosophila* leveraging molecular cloning, transgenic, and fluorescence microscopy techniques.

Summary of Skills
- Extensive experience with high performance and cloud computing (UGE and AWS) for analysis of genomic (variant discovery and assembly), transcriptomic (assembly, quantification, isoform detection), and proteomic (alignment and orthology) datasets.
- Excellent knowledge and experience in R and Python for modeling of large datasets.
- Proficient in computational workflow development with Snakemake, Argo, and Databricks.
- Knowledgeable in software engineering concepts such as continuous integration and deployment, version control (git), testing (unit and integration), and Docker.
- Proficient in high throughput CRISPR guide RNA design and construct cloning.
- Experienced in DNA extraction methods for large-scale genotyping, next generation sequencing, and high molecular weight DNA for third generation sequencers (PacBio, Nanopore).
- Highly experienced in RNA extraction and cDNA preparation for allele specific expression assays.
- Nine years of experience managing diversity and mapping populations for quantitative, genomic, and reverse genetic analyses.
- Proficient with dissection for stereo and fluorescence microscopy.
- Standard laboratory skills: pipetting, PCR, electrophoresis, and bacterial/cell culture.
- Project leadership, scientific communication, line management, and project management.

Publications

Kim Y, Kim GW, Han K, Lee HY, Jo J, Kwon JK, **Lemmon ZH**, et al. (2022) Identification of Genetic Factors Controlling the Formation of Multiple Flowers Per Node in Pepper (*Capsicum* spp.) Frontiers in Plant Science 13, 884338. doi: 10.3389/fpls.2022.884338

Ranallo-Benavidez TR, **Lemmon ZH** et al. (2021) Optimized sample selection for cost-efficient long-read population sequencing. Genome Res 31(5), 910-918. doi: 10.1101/gr.264879.120

Alonge M, Benoit M, Van Der Knaap E, Schatz M, Lippman Z, Soyk S, Pereira L, Zhang L, **Lemmon ZH** et al. (2020) Major Impacts of Widespread Structural Variation on Gene Expression and Crop Improvement in Tomato. Cell 182, 1–17. doi: 10.1016/j.cell.2020.05.021

Rodriguez-Leal D, Xu C, Kwon C, Soyars C, Demesa-Arevalo E, Man J, Liu L, **Lemmon ZH**, Jones D, Van Eck J, Jackson D, Bartlett M, Nimchuk Z, Lippman Z (2019) Evolution of buffering in a genetic circuit controlling plant stem cell proliferation. Nat. Genet. 51, 1 (2019). doi: 10.1038/s41588-019-0389-8

Lemmon ZH, Reem N, Dalrymple J, Soyk S, Swartwood K, Rodriguez-Leal D, Van Eck J, Lippman ZB (2018) Rapid improvement of domestication traits in an orphan crop by genome editing. Nat. Plants 4, 766–770. doi: 10.1038/s41477-018-0259-x

Wang X, Chen Q, Wu Y, **Lemmon ZH**, et al (2017) Genome-wide analysis of transcriptional variability in a large maize-teosinte population. Mol. Plant. doi:10.1016/j.molp.2017.12.011

Rodríguez-Leal D, **Lemmon ZH**, Man J, Bartlett ME, and Lippman ZB (2017) Engineering Quantitative Trait Variation for Crop Improvement by Genome Editing. Cell 1–11. doi:10.1016/j.cell.2017.08.030

Soyk S, **Lemmon ZH**, Oved M, Fisher J, Liberatore KL, Park SJ, Goren A, Jiang K, Ramos A, van der Knaap E,Van Eck J, Zamir D, Eshed Y, Lippman ZB (2017) Bypassing Negative Epistasis on Yield in Tomato Imposed by a Domestication Gene. Cell 169, 1142–1155.e12. doi:10.1016/j.cell.2017.04.032

Lemmon ZH, Park SJ, Jiang K, Van Eck J, Schatz MC, Lippman ZB (2016) The Evolution of Inflorescence Diversity in the Nightshades and Heterochrony during Meristem Maturation. Genome Res 26(12): 1676-1686. doi:10.1101/gr.207837.116.

Lemmon ZH, Bukowski R, Sun Q, Doebley J (2014) The Role of *cis* Regulatory Evolution in Maize Domestication. PLoS Genet 10: e1004745. doi:10.1371/journal.pgen.1004745.

Lemmon ZH, Doebley J (2014) Genetic Dissection of a Genomic Region with Pleiotropic Effects on Domestication Traits in Maize Reveals Multiple Linked QTL. Genetics 198: 345–353. doi:10.1534/genetics.114.165845.

Lang Z, Wills DM, **Lemmon ZH**, Shannon LM, Bukowski R, Wu Y, Messing J, Doebley J (2014) Defining the Role of *prolamin-box binding factor1* Gene During Maize Domestication. J Hered. doi:10.1093/jhered/esu019.

Patents

Gault CM, **Lemmon ZH**, et al. 2024. Soybeans with combinations of gene mutations. U.S. Patent 20,240,301,012, filed March 15, 2024, and issued September 12, 2024.

Kremling K, Goff S, Mei W, Li R, Altman R, **Lemmon ZH**. 2022. Methods and systems for assessing genetic variants. U.S. Patent 20,220,277,807, filed February 17, 2022, and issued September 1, 2022.

Lippman ZB, Soyk S, **Lemmon ZH**. 2022. Mutations in mads-box genes and uses thereof. U.S. Patent 20,220,002,740, filed November 14, 2019, and issued January 6, 2022.

Conference Abstracts

Lemmon ZH, Park SJ, Jiang K, Van Eck J, Schatz MC, Lippman ZB. The Evolution of Inflorescence Diversity in the Nightshades and Heterochrony during Meristem Maturation. Poster session presented at: International Plant and Animal Genome XXV. January 14-18, 2017. San Diego, CA, USA.

Lemmon ZH, Park SJ, Jiang K, Van Eck J, Schatz MC, Lippman ZB. The Evolution of Inflorescence Diversity in the Nightshades and Heterochrony during Meristem Maturation. Talk presented at: The 13th Solanaceae Conference. September 12-16, 2016. Davis, CA, USA.

Lemmon, ZH, Park SJ, Jiang K, Schatz MC, Lippman ZB. Gene expression divergence during reproductive transition drives Solanaceae inflorescence architecture diversity. Poster session presented at: FASEB: Mechanisms in Plant Development. August 2-7, 2015. Saxton River, VT, USA.

Lemmon ZH, Bukowski R, Sun Q, Doebley J. Gene Regulatory Change in Maize Domestication. Poster session presented at: 55th Annual Maize Genetics Conference. 2013 March 14-17. St. Charles, IL, USA.

Lemmon ZH, Bukowski R, Sun Q, Doebley J. Genome-Wide Allele Specific Expression Assays in Maize-Teosinte Hybrids. Poster session presented at: 54th Annual Maize Genetics Conference. 2012 March 15-18. Portland, OR, USA.

Lemmon ZH, Doebley J. Characterization of Domestication QTL on Chromosome Five of *Z. mays*. Poster session presented at: 53th Annual Maize Genetics Conference. 2011 March 17-20. St. Charles, IL, USA.

Lemmon ZH, O'Tousa J. Characterization of Apoptosis Inducing Factor (AIF) in the *Drosophila* visual system. Poster session presented at: 49th Annual Drosophila Research Conference. 2008 April 2-6. San Diego, CA, USA.

Synergistic Activities

Undergraduate Research Program Mentor — Summer 2017
Cold Spring Harbor Laboratory, Cold Spring Harbor, NY

Teaching Assistant, Genomics — Fall 2014
Cold Spring Harbor Laboratory, Cold Spring Harbor, NY

Genetics Curriculum Committee — January 2012 – May 2014
University of Wisconsin, Laboratory of Genetics, Madison WI

Genetics Written Preliminary Exam Committee — April 2011 – July 2011
University of Wisconsin, Laboratory of Genetics, Madison WI

Graduate Teaching Assistant — January 2010 – May 2010
University of Wisconsin, Laboratory of Genetics, Madison WI

Undergraduate Mentoring (multiple students) — May 2009 – December 2012
University of Wisconsin, Laboratory of Genetics, Madison WI

Undergraduate Teaching Assistant — Fall 2006 & 2007
University of Notre Dame, Department of Biology, Notre Dame IN